

# LAN Topologies: Bus, Star, Ring, and Mesh

Topology describes how nodes of a local area network are arranged. A bus topology places every station on a single coaxial cable. Terminators absorb the signal at both ends. The design is inexpensive, but a cable break disables the entire segment and troubleshooting is difficult.

A star topology connects each workstation to a central hub or switch. This is the default for twisted-pair Ethernet. A failed cable affects only one host, and moves are simple: unplug a patch cord and plug it into another switch port. The central device becomes a critical point of failure, so important closets use redundant switches and dual power supplies.

Ring topologies pass a token or follow a logical ring, as in legacy Token Ring and some metropolitan fiber rings. Mesh topologies provide multiple paths and are used between core switches. A hybrid campus LAN usually combines a star at the access layer with a partial mesh in the core.

Choosing a topology is a reliability decision as much as a cabling decision. Star wiring with hierarchical switching scales from a single laboratory to a multi-floor building. Mesh uplinks protect against a single backbone failure. Documentation of the physical topology should match the logical drawing that operators keep in the network operations center.